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Phenological normalization can improve in-season classification of maize and soybean: A case study in the central US Corn Belt

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ABSTRACT

Within-season estimates of crop-specific planted area, conditions, and expected yields are critical for decision- and policy-making related to agriculture and food security. However, these estimates require in-season crop type maps which are not currently widely available. Machine learning and remote sensing data can be used to create crop type maps that provide crop-specific land cover classifications to enable timely analysis at field scales throughout the growing season to complement traditional reporting efforts. However, existing methods are often limited by lower performance on test data from seasons with different patterns not seen during training and inability to provide classifications during the growing season when most operationally relevant. We present a new approach to in-season crop type mapping that addresses inter-annual domain shift by normalizing satellite observations by land surface phenology stage. These phenology-normalized observations are input to a neural network that extracts temporal and spatial features using both recurrent and convolutional layers respectively. Using Harmonized Landsat and Sentinel-2 (HLS) and Sentinel-1 SAR observations with test data from the U.S. states of Iowa in 2019 (late planting year) and Illinois in 2020 (standard year), we showed that this method enabled good performance in both scenarios throughout the growing season (71% and 72% in early-season, 80% and 84% in mid-season, 81% and 85% in late-season for Iowa 2019 and Illinois 2020 respectively).

1. Introduction

Within-season estimates of crop-specific planted area, conditions, and expected yields are critical for decision- and policy-making related to agriculture and food security. Important decisions are made based on crop production estimates around the clock by different actors in the commodity markets. Official reports like the USDA's Crop Progress and Condition Reports are widely followed by market analysts and traders to inform decisions about agricultural markets and futures. Thus, uncertainty in public information including official reports for major food-producing countries can result in uncertainty and volatility in agricultural markets and prices (Lehecka, 2014). This uncertainty is exacerbated when shocks occur such as extreme weather events (LeComte, 2020; Hosseini et al., 2020; Ajadi et al., 2020), pest or disease infiltration, economic sanctions/tariffs, or armed conflict.

Information about global crop progress and conditions derived from Earth observations (EO) data can enable timely analysis during the growing season to complement existing data collection and reporting efforts (Becker-Reshef et al., 2019a). Methods for estimating key agricultural quantities such as planted acreage, production, condition, and yield require in-season crop type maps as inputs, yet in-season crop type maps are not currently available to the public. Prior work on crop type classification has demonstrated promising results using time series observations to differentiate spectrally similar crop types. However, models trained to classify crop types based on temporal patterns may not generalize well to inputs from years that have a different distribution than in training years, e.g., due to changes in planting timelines or climate change. For example, in spring 2019, heavy snowfall and rains caused unprecedented flooding across the Midwestern states and forced farmers to plant crops as much as 1–2 months late in some places (Irwin and Hubbs, 2019). Such extreme weather events may exacerbate uncertainty and market volatility. For crop type classification models to be used operationally, it is important that they 1) have robust performance for seasons that differ from those seen in training, and 2) provide

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predictions within the growing season when uncertainty is higher.

Most prior approaches to crop type classification include observations from multiple spectral bands and time steps in the classifier input. Random forests and decision trees have shown good performance in many case studies and thus are commonly used for crop type and land cover classification (e.g., Hariharan et al. (2018); Neetu and Ray (2019); Palchowdhuri et al. (2018); Saini and Ghosh (2018); Shelestov et al. (2019); Shukla et al. (2018); Song et al. (2017); Sonobe et al. (2017a,b, 2018); Viskovic et al. (2019); Wang et al. (2019a,b, 2020); Nguyen et al. (2020)). For example, Song et al. (2017) used bagged decision trees to classify time series statistics from Landsat spectral bands, whereas Wang et al. (2019b) and Wang et al. (2020) used the coefficients of harmonic regression as input features for a random forest. Nguyen et al. (2020) derived phenological metrics from a fitted quadratic model of enhanced vegetation index (EVI) as a function of accumulated growing degree days (AGDD), and used these metrics as inputs features to a random forest classifier of land cover and crop type classes. Kernel methods including support vector machines (SVMs) and kernel-based extreme learning machines (Pal et al., 2013) have also shown good performance for crop type classification (e.g., Feng et al. (2019); Neetu and Ray (2019); Saini and Ghosh (2018); Sonobe et al. (2017a,b); Wagstaff and Mazzoni (2006); Wang et al. (2019a)). For example, Feng et al. (2019) extracted more than 100 features including spectral, textural, and phenological properties of optical observations for SVM classification.

However, such methods often require domain-specific, hand-crafted input features to achieve good classification performance. Recent work demonstrated improved classification performance using neural network-based approaches that automatically learn useful features for discriminating between target classes based on spatial patterns (using convolutional neural networks) and temporal patterns (using recurrent neural networks). Zhong et al. (2019) showed that a convolutional neural network (CNN) with 1D convolutions outperformed a long short term memory (LSTM) network (Hochreiter and Schmidhuber, 1997) for classifying time series inputs of enhanced vegetation index (EVI). Cai et al. (2018) showed that better performance could be obtained for deep neural networks when time series inputs include several spectral bands and indices compared to just one. Xu et al. (2020) showed that an LSTM with bidirectional flows and attention mechanisms improved crop type classification performance. Kussul et al. (2017) found that ensemble of 2D CNNs had better performance compared to 1D CNNs with optical and radar time series inputs. Ruβwurm and Korner (2017) showed that an LSTM gave better performance than a 2D-CNN for multispectral time series inputs that also included a neighborhood of pixels around the predicted pixel. Similarly, Garnot et al. (2019) found that temporal features extracted by recurrent layers contributed more to crop type classification accuracy than spatial features extracted by 2D convolutional layers, but that the greatest performance was achieved by combining both recurrent and convolutional layers. Brandt (2019) showed that a network with capsule and attention layers in addition to CNN and LSTM layers performed better for crop type classification than a 2D CNN, LSTM, and combined CNN-LSTM. Ji et al. (2018) showed that 3D CNNs which performed operations on the spatial, spectral, and temporal dimensions simultaneously sometimes improved performance over compared methods. Gray et al. (2021) showed that using both convolutional and recurrent layers gives better land cover classification performance than those that use only convolutional or recurrent layers. Together, these studies suggest that better crop type classification performance can be achieved with deep learning methods that combine both recurrent and convolutional layers to learn relevant patterns in spectral, spatial, and temporal dimensions of remote sensing observations.

To be useful in an operational setting, crop type classification methods must be able to be applied during the growing season, yet relatively few studies account for partially-observed time series during the growing season and assess in-season classification performance (e.g., Zhong et al. (2016); Dahal et al. (2018); Cai et al. (2018); Xu et al. (2020); Konduri et al. (2020); Johnson and Mueller (2021)). The USDA Cropland Data Layer (CDL) provides annual crop type maps with 30 m/pixel resolution for the continental United States based on satellite observations. While this product is widely used and has good accuracy for corn and soybeans in the Midwestern US states, it is not available during the growing season and relies on a large training dataset of ground-truth samples collected during the growing season each year Boryan et al. (2011). Furthermore, few studies have assessed model generalization to different years or regions that exhibit growing season timelines that differ from those used in training (e.g., Xu et al. (2020); Konduri et al. (2020); Wang et al. (2020); Ajadi et al. (2021); Zhang et al. (2021)).

This work aims to provide a computationally feasible and affordable solution for producing in-season crop type maps for major crops (corn and soybeans) in the United States, which are not currently available to the public. We present a method for crop type classification that uses time series inputs at three land surface phenology stages detected at the pixel level using Harmonized Landsat and Sentinel-2 (HLS) optical and Sentinel-1 synthetic aperture radar (SAR) observations to mitigate the effects of domain shift caused by differences in planting timelines. This method enables in-season predictions of crop type at three time periods during the growing season (early, middle, and late season). We evaluated the performance of this method for classifying corn, soybean, and other land cover in a study area of the central US Corn Belt within the growing seasons of 2019 and 2020.

This study addresses the following research questions:

- What is the best approach for selecting training data in terms of geographic diversity for operational crop mapping using moderate-resolution (30 m) satellite imagery?
- How can we prevent reduced accuracy for unseen years when there is a domain shift with respect to prior years (e.g., due to late planting)?
- Which crop type classification methods perform best throughout the growing season?

We provide open code and maps to promote future research and operational use. The data can be found at https://data.harvestportal.org/dataset/us-in-season-croptype-maps and the code at https://github.com/nasaharvest/us-inseason-phenorm.

2. Datasets

2.1. Labeled data

The Cropland Data Layer (CDL) is released several months after each growing season by the United States Department of Agriculture (USDA) (Boryan et al., 2011) and provides classifications of crop type and other land cover at 30 m/pixel resolution. CDL has been used as a surrogate for ground truth data in prior studies since it is based on extensive survey data collected by the USDA (e.g., Chakrabarti et al. (2019); Wang et al. (2019b); Xu et al. (2020)) and typically has high accuracy for corn and soybeans in midwestern states (Boryan and Yang, 2013). Lark et al. (2017) recommends aggregating classes and filtering within-field speckle to reduce errors in CDL labels. We re-classified the CDL into three classes: corn, soybeans, and "other". The corn and soybean classes correspond to the same classes in the CDL. The "other" class included all other classes, including crops other than corn and soybeans (e.g., alfalfa or wheat) and non-crop classes (e.g., pasture or forest). We used these re-classified CDL labels for this study. To reduce the likelihood of mis-classified samples resulting from within-field speckle, we applied a 3×3 homogeneity filter to identify non-homogeneous regions which were not used in the training set. We did not use this mask for computing evaluation metrics on the validation or test sets.

In addition to CDL, we obtained a crop type reference dataset from commercial company Intelinair to evaluate our in-season 2020 classification results. Intelinair collects high resolution (10 cm/pixel) aerial

images of agricultural areas throughout the growing season to provide insights to growers. Intelinair created a reference dataset of 3973 fields in Illinois and Indiana by visually interpreting the crop type in images acquired during the 2020 corn and soybean growing season. Intelinair provided this dataset for our study as a set of points located in the center of each field with an associated crop type label of corn, soybean, or other crop. Each field was reviewed by multiple Intelinair labelers. Of the total 3973 fields, 2478 (62%) are corn, 1303 (33%) are soybean, and 192 (5%) are another crop. Fig. 1 shows the spatial distribution of the field center locations.

2.2. Satellite data

We used Harmonized Landsat-8 and Sentinel-2 (HLS) optical images and Sentinel-1 synthetic aperture radar (SAR) observations to construct the model input. HLS harmonizes Sentinel-2 and Landsat-8 surface reflectance products to enable the satellites to be treated as a virtual constellation with 30 m/pixel spatial resolution and 2–3 days revisit time; products are available within 2–10 days of image acquisition (Claverie et al., 2018). We used observations acquired between 2017 and 2020 since the second of the two Sentinel-2 satellites was not launched until 2017. We masked out pixels with clouds and cloud shadows (bit numbers 0–3) according to the HLS quality assessment (QA) layer. We also used observations from the Sentinel-1 satellite, which acquires C-band synthetic aperture radar (SAR) observations every 6 days at the equator. We included Sentinel-1 SAR observations in addition to the HLS optical observations because prior work has shown that use of both datasets in the input to land cover classification models improves model performance (e.g., Orynbaikyzy et al. (2019); Kenduiywo et al. (2017); Kussul et al. (2017); Hariharan et al. (2018); Sonobe et al. (2017b)). We used the VV and VH polarisations as in most prior studies.

This study focuses on a region spanning multiple states in the central U.S. Corn Belt. We used HLS tiles that intersected with the Iowa, Illinois, or the Intelinair reference points, resulting in 42 HLS tiles that cover all of Iowa and Illinois, more than half of Indiana, and small regions of bordering states (South Dakota, Nebraska, Kansas, Minnesota, Missouri, and Wisconsin). The tile footprints and IDs are shown in Fig. 1. We downloaded HLS observations via HTTPS (release version 1.4) (https://hls.gsfc.nasa.gov/). We downloaded Sentinel-1 observations from the Sentinel-1 Radiometric Terrain Corrected (RTC) dataset hosted on Amazon Web Services (AWS) (https://registry.opendata.aws/sentinel-1-rtc-indigo/). This dataset was processed by Indigo Ag; pre-processing steps include re-gridding to the MGRS tile system used by HLS, speckle filtering, and terrain correction.

3. Methods

3.1. Phenology normalization

Crops undergo specific crop growth or phenological stages that vary between individual crops. While these granular crop phenology stages are difficult to detect at the spatial and temporal resolution of most satellite datasets, cropped pixels in satellite images typically exhibit three key stages in the *land surface* phenology that can be observed using remote sensing: greenup, when crops emerge from the ground and undergo early vegetation growth; peak growth, when crops reach their maximum growth and begin to ripen; and senescence, when crops dry down before they are harvested. For many crops including corn and soybeans, these land surface phenology stages are apparent in the time series of vegetation indices like normalized difference vegetation index (NDVI, e.g., Gao et al. (2020); Becker-Reshef et al. (2019b); Wardlow et al. (2006)), though regional planting timelines may vary from year to year due to weather or other climate-related factors that influence soil conditions or temperature (Hatfield and Prueger, 2015). For this reason, it is important that observations are compared at the same phenological

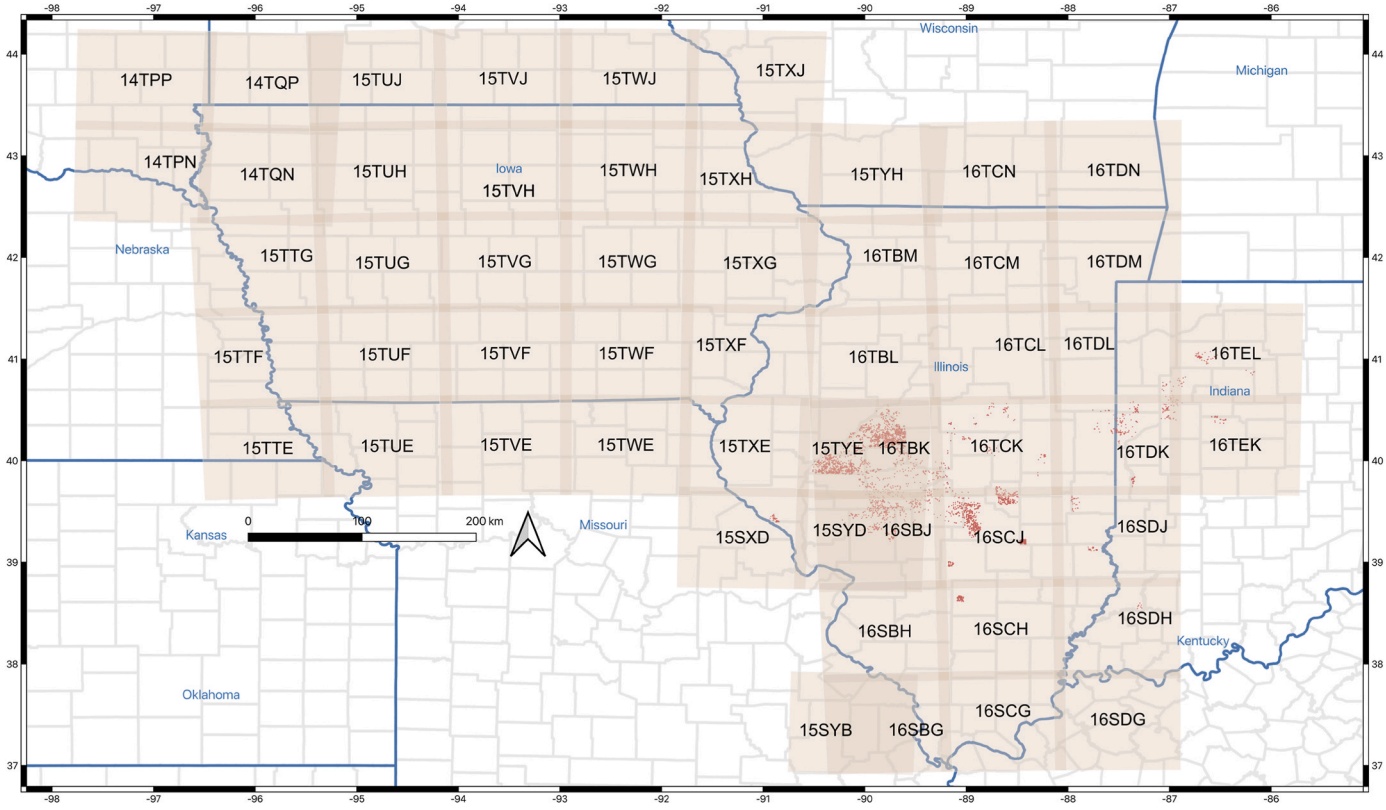


Fig. 1. HLS (MGRS grid system) tiles used for this study. Red points show locations of Intelinair reference labels. State borders and names are in blue. Gray lines show county borders. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

stage, which may not be at the same time (Skakun et al., 2019). Satellite observations of the same crop type on the same day of two different years will be less comparable if the time at which the crops were planted differs between the years, causing the growth signature to be shifted in the time domain. Xu et al. (2020) described this as the reason for poor generalization of their crop type classifier to regions where crop growth progress at the same week differed from the training data. The top row in Fig. 2 shows the gap-filled HLS false color (R = SWIR, G = NIR, B = red) image for a region of cropland in northern Illinois on approximately the same date in 2017, 2018, and 2019. The crops in this image appear to be at different land surface phenology stages despite being taken at the same time of the season—the level of vegetation appears especially lower in 2019 than prior years due to widespread planting delays.

To reduce differences in remote sensing observations of crops in different years that are due to planting timelines, we propose an approach that classifies satellite observations of crops at the same land surface phenology stage rather than at the same dates. In other words, we transform the remote sensing time series defined by dates to a time series defined by land surface phenology; this can be considered a type of domain adaptation (Tuia et al., 2016). Fig. 3 illustrates the steps for performing phenology normalization. For each pixel in our dataset, we first resampled the HLS time series to 5-day intervals and performed linear interpolation to fill gaps resulting from clouds or other missing data sources. We then computed the approximate date of greenup, peak growth, and senescence using the resampled, gap-filled NDVI time series. We defined the greenup DOY to be the day between April 15 and July 1 after which the greatest increase in NDVI was observed between consecutive observations. We defined the peak DOY to be the day after greenup but before September 30 at which the highest NDVI was observed between consecutive observations. We defined the DOY of senescence to be the day, following the peak DOY, after which the greatest decrease in NDVI was observed between consecutive observations. Since this detection is based on slopes between timesteps bounded by seasonal timeframes, it does not require the complete time series for the season and can be applied during the growing season. We then selected the HLS reflectance values from the time series of each band for these three phenological stages to construct a "phenology-normalized" multi-spectral image with three time-steps. In each pixel, we selected the value from the nearest (in time) available Sentinel-1 SAR observation to the phenology dates. Fig. 2 illustrates the effect of phenology normalization. In the top row, pixel values are all from the same date in each image. Despite each image being acquired on approximately the same day, there are large differences in the reflectance values due to inter-annual differences in crop progress, especially for 2019. The bottom row of Fig. 2 shows the same regions after phenology normalization, in which each pixel shows the reflectance values at the same phenological stage (peak NDVI) rather than the same date. After phenology normalization, the inter-annual differences in the distribution of reflectance values is substantially reduced compared to the top row. Note that the artifacts in the bottom row images are mainly due to cloud shadows not detected by the HLS cloud mask; these artifacts are less prominent in the top row because the images were smoothed for visualization purposes using the Savitzky-Golay algorithm (Chen et al., 2004) (which is less sensitive to un-detected clouds), whereas we used

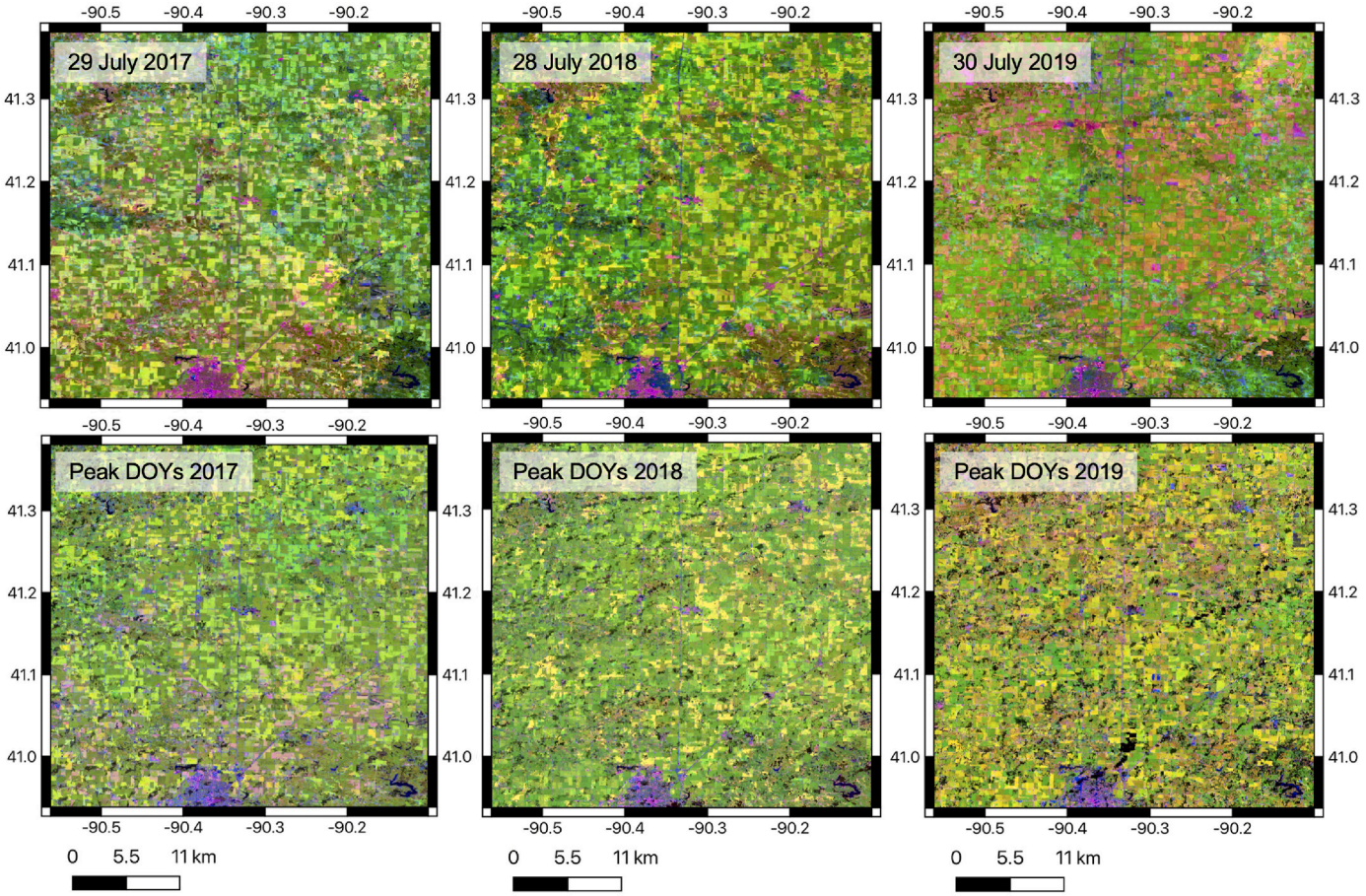


Fig. 2. False color images (R = SWIR, G = NIR, B = red) before (top row) and after (bottom row) phenology normalization. In the top row, pixel values are all from the same date in each image while in the bottom row each pixel value is from the date detected as the day of year (DOY) with peak NDVI. Phenology normalization reduces differences in the appearance in crops that are due to inter-annual differences such as planting timelines, thus images in the bottom row appear more uniform compared to the top row. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

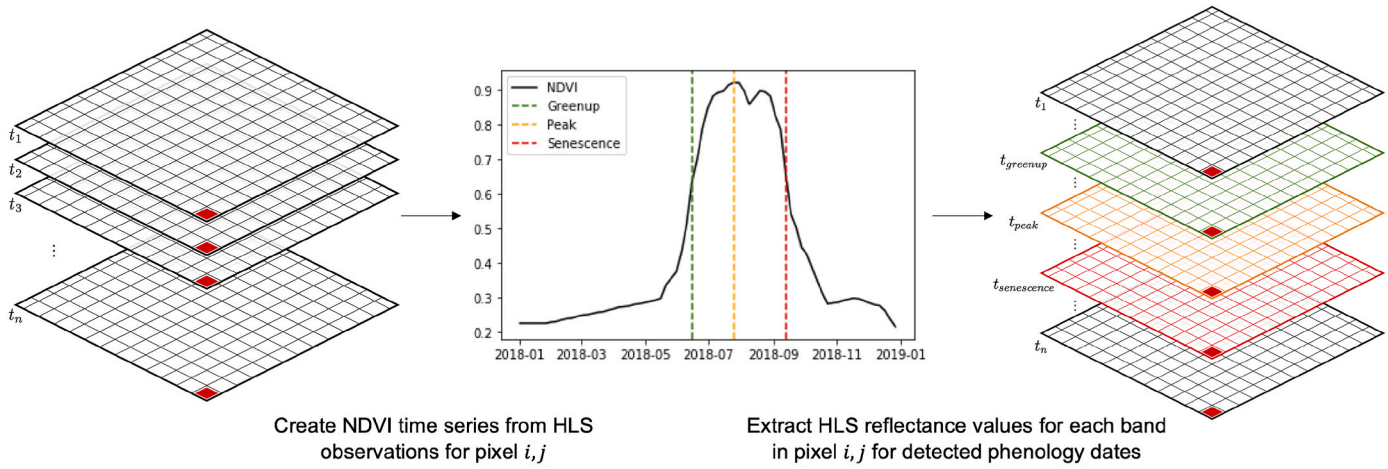


Fig. 3. Land surface phenology normalization steps.

linear interpolation (which is much faster than Savitzky-Golay) in the phenology normalization pre-processing.

3.2. Classification model

We used the blue, green, red, near infrared (NIR), and short-wave infrared (SWIR) HLS bands (Landsat-8 bands B02–B07; Sentinel-2 bands B02, 03, 04, 08A, 11, 12) and the VV and VH polarisations from Sentinel-1 to form the classifier input. We also included two spectral indices, normalized difference water index (NDWI) and land surface water index (LSWI), since these have been shown to be useful for distinguishing corn and soybeans by capturing water content inside the leaf (Cai et al., 2018; Feng et al., 2019; Xu et al., 2020). We calculated NDWI and LSWI as:

$$\text{NDWI} = \frac{\text{GREEN} - \text{SWIR1}}{\text{GREEN} + \text{SWIR1}} \tag{1}$$

$$\text{LSWI} = \frac{\text{NIR} - \text{SWIR1}}{\text{NIR} + \text{SWIR1}} \tag{2}$$

In our experiments we found that including NDWI and LSWI in the input had better classification performance than without.

Our model architecture (Fig. 4) consists of two branches: a 2D CNN to capture spatial and/or textural patterns and an LSTM to capture temporal patterns. The CNN and LSTM outputs are joined by a dense layer and then passed through the final softmax prediction layer. Since the number of growing days may be a useful distinguishing factor between corn and soybeans, we also included the scalar number of days elapsed between the senescence DOY and greenup DOY as an input feature, i.e.: $\delta = \text{DOY}_{\text{senescence}} - \text{DOY}_{\text{greenup}}$. This feature is input to the dense layer that merges the CNN and LSTM outputs. For the mid-season classification when only the greenup and peak inputs are available to the model, the elapsed time is between the greenup and peak DOY. The δ feature is not used in the early-season classification since only one timestep is available.

The first layer in the recurrent branch is a dense linear activation layer with 16 units followed by four LSTM layers with 16 units in each layer. Each LSTM layer is followed by a batch normalization (Ioffe and Szegedy, 2015) layer and dropout (Srivastava et al., 2014) layer (randomly dropping 30% of units). We used the rectified linear unit (ReLU) activation function for all layers. The inputs to the LSTM branch are the phenology-normalized reflectance values for three timesteps

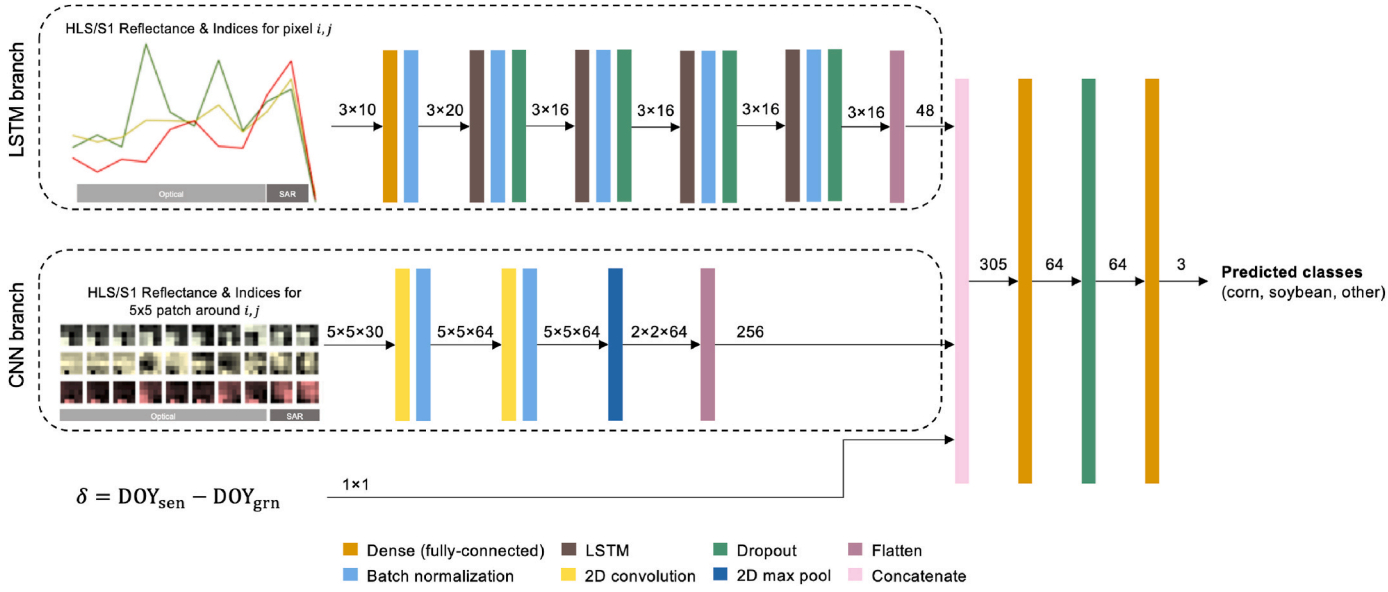


Fig. 4. Model architecture for in-season crop type classification. Input and output dimensions for each layer are shown above arrows (not including the batch size dimension).

(greenup, peak, and senescence) in one pixel location. This includes 6 HLS reflectance values, 2 HLS spectral index values, and 2 SAR values for each timestep—i.e., $x_i^{\text{LSTM}} \in \mathbb{R}^{M \times D}$ where $M = 10$ is the number of channels and $D = 3$ is the number of timesteps.

The CNN branch has two 2D convolution layers. We found that validation accuracy did not improve when fewer or more layers were used. Each layer has 64 filters, 3 × 3 kernels, and 1-pixel strides. We used ReLU activation functions and batch normalization after each layer. We found that dropout in the CNN layers did not improve validation accuracy. Each convolution layer was followed by a 2 × 2 max pooling layer. The CNN branch input consists of 5 × 5-pixel patches around the pixel to be classified. We experimented with different patch sizes and found that larger patch sizes resulted in over-segmentation of crop fields while smaller patch sizes caused more in-field speckle errors (Fig. 5). We concatenated the HLS and S1 patches for each timestep, resulting in an input tensor of shape $x_i^{\text{CNN}} \in \mathbb{R}^{k \times k \times MD}$ where $k = 5$ is the input patch size, $M = 10$ is the number of bands, and $D = 3$ is the number of timesteps.

The outputs from the CNN and LSTM branches are flattened before being input to a dense (fully-connected) layer with 64 units and dropout. This dense layer is followed by the final softmax output layer that represents the posterior probability of an input belonging to the *corn* (0), *soybean* (1), or *other* (2) class. This model can be used for in-season crop type classification at three stages during the growing season. Early in the season (April 15-June 30), only the first timestep (greenup) is included in the model input. In the middle of the season (July 1-September 30), the first two timesteps are included in the input. All three timesteps are included in the late-season classification which can be run in November. We implemented the models using the Keras deep learning library with TensorFlow backend.

4. Experiment setup

We conducted three experiments to address the research questions outlined in Section 1. In all experiments, we standardized the input data by subtracting the mean and dividing by the standard deviation computed band-wise from the training dataset.

4.1. Geographic partitioning of training data

Satellite datasets are typically organized into tiles based on a grid system, such as the Military Gridded Reference System (MGRS) used by HLS. These tiles are very large compared to typical machine learning model input data sizes: one HLS tile with 30 m/pixel resolution spans 109.8 km × 109.8 km or 3, 660 × 3, 660 pixels and has many spectral bands and time steps. Thus a single tile is often treated as a data set in itself (e.g., Xu et al. (2020)). While this approach can be straightforward to implement, it may exclude broader trends in agriculture across multiple tiles or the surrounding region that could enable more accurate classification. We propose multiple approaches for spatial data partitions used for model training and performed experiments to identify which approach gives the best overall map accuracy while promoting simplicity and avoiding overfitting. These approaches, visualized in Fig. 6, are:

1. Local model: a separate model trained for each tile with training data from that tile.
2. Regional model: a separate model trained for each tile with training data from that tile and all adjacent tiles.
3. Universal model: a single model with training data from all tiles.
4. Agro-ecological cluster model: a separate model trained using data from each of *N* agro-ecological clusters. We determined the clusters using *k*-means clustering with 5 clusters (Likas et al., 2003) using spatial datasets of agro-ecological variables that might influence agricultural practices and patterns. We used the USGS National Elevation dataset for elevation and the Famine Early Warning System Network (FEWS-Net) for humidity, rainfall, air temperature, surface temperature, soil temperature, and soil humidity (McNally et al. (2017)) for clustering.
5. Ensemble model: a model combining the predictions from all four models using a voting classifier. The voting classifier selects the label chosen most frequently across the four models. If there is a tie, the label with the lowest integer value is selected (e.g., corn would be chosen in a tie-break between corn and soybeans since they have integer labels 0 and 1 respectively.).

For all approaches, we performed inference on a per-tile basis to enable direct comparisons of methods at the tile level (Table 1). We used a baseline Random Forest classifier for this experiment because this method is widely used in the crop type and land cover classification literature and we wanted to ensure our findings would be broadly applicable, though we expect the relative performance across approaches would hold for other methods. We performed this experiment using all tiles intersecting Iowa. The model input was the same as the late-season CNN-LSTM model, with spatial and spectral dimensions flattened since Random Forests require flattened feature vectors (Breiman, 2001). We used the default parameters for the random forest classifier in Scikit-learn (Pedregosa et al., 2011). The number of training samples was fixed for all models (100,000 samples balanced equally between corn, soybean, and other). We did not apply any post-classification despeckling as described for later experiments.

4.2. Comparison to Cropland Data Layer

Domain adaptation techniques are particularly important for improving model robustness to seasons in which the distribution of the remote sensing inputs differs substantially from the training data. The 2019 growing season in the US was characterized by widespread planting delays due to flooding and excess moisture in the spring planting months, which translated to shifts in the data distributions



Fig. 5. Effect of CNN input patch size on final classification: larger patches result in over-segmentation while smaller patches result in speckle errors.

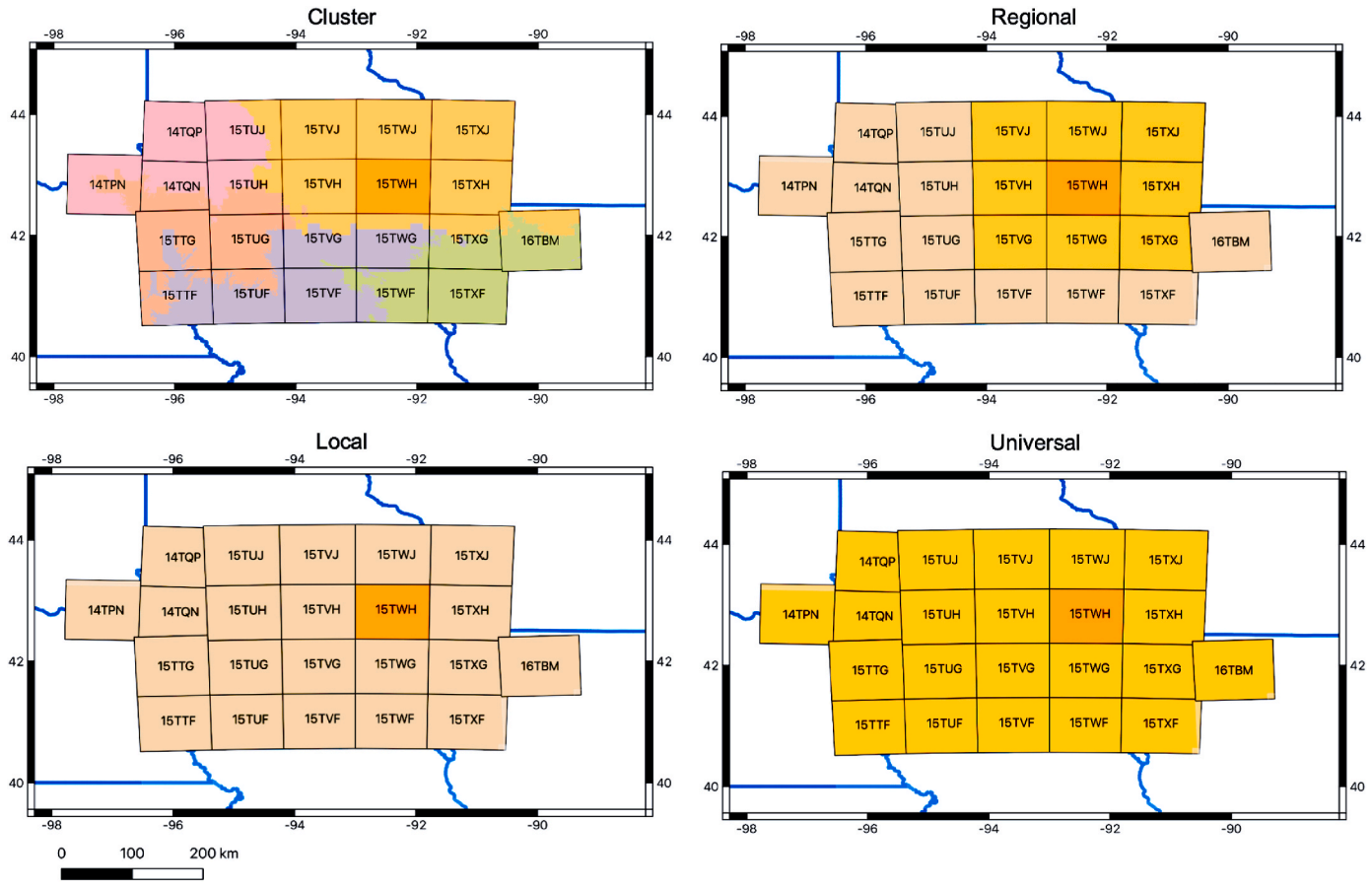


Fig. 6. Approaches for partitioning training data/models. In this example, the bright orange tile—15TWH—is the test tile. The surrounding orange tiles indicate from which additional tiles/regions training pixels are sampled under each approach for that test tile. In the local model, training pixels are sampled from only one tile (the test tile) whereas in the universal model, training pixels are sampled from all tiles. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

between 2019 and prior years. To evaluate the ability of the proposed method to predict crop types despite this domain shift, we used data from 2017 and 2018 for constructing the training set, 2018 for the validation set, and 2019 for the test set, using all HLS tiles in Iowa. We used the CDL labels (aggregated to three classes: corn, soybean, other) as a surrogate for ground-truth labels. We performed three experiments to evaluate the crop type classification accuracy when the model is given one (greenup only), two (greenup and peak), or three (greenup, peak, and senescence) timesteps to measure performance at different land surface phenology stages within the growing season. To evaluate the performance of the model in a non-anomalous year, we performed the same experiments using data from 2017 to 2019 for training, 2019 for validation, and 2020 for testing, this time using all HLS tiles in Illinois. In addition to the proposed neural network classifiers, we evaluated a baseline random forest classifier with default parameters from Scikit-learn.

We used 100,000 training points and 10,000 validation points for each model. We used the regional approach described in Section 5.1: we trained a separate model for each HLS tile using training samples from that tile and adjacent tiles. We randomly sampled points from the tiles such that there was an equal number of training points from each tile and the three output classes were balanced, using a random seed of 488. We used early stopping to determine how long to train each model so that training was automatically terminated if the validation loss did not improve for 9 epochs (passes through the training data set). We used a batch size of 512 and Adam optimizer with default hyperparameters to train the models (Kingma and Ba, 2014). After performing inference for all tiles, we used GDAL to mosaic the tile-wise outputs and clip the merged raster to the state border. Finally, we used GDAL to apply a sieve filter to reduce in-field speckle errors for all predicted maps (results are reported after despeckling).

4.3. Comparison to in-season reference dataset

We used the same experimental setup as in the previous section to train each model using data from 2017 to 2019 for training, 2019 for validation, and 2020 for testing. We used the 3973 samples (62% corn, 33% soybean, 5% other crops) in the Intelinair reference dataset described in Section 2 for testing. As in prior experiments, we evaluated a random forest in addition to our proposed models and reported results after despeckling.

5. Results

5.1. Geographic partitioning of training data

Of the four approaches (excluding the ensemble), we found that the regional model had the highest accuracy and was the most consistent (lowest range) across tiles. The cluster model had the highest accuracy in 8 of the tiles but lower mean accuracy than the regional model. The local model had the lowest mean accuracy and was the least accurate for most tiles. The ensemble model had a higher mean accuracy than all of the individual models, but at the tile level had the highest accuracy for only about half of the tiles.

Table 1
Overall accuracy (%) for HLS tiles in Iowa for each partitioning approach, using CDL 2019 as labels. **Bold** indicates best performance. Range column indicates difference between maximum and minimum score across the four partitioning approaches for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile	Local	Universal	Clusters	Regional	Range	Ensemble
14TPN	76.4	77.5	71.1	77.1	7.4	**78.5**
14TQN	77.5	**80.3**	79.9	80.0	2.8	**80.3**
14TQP	70.8	73.1	**75.1**	73.4	4.3	73.3
15TTG	73.1	74.1	73.6	71.8	2.4	**74.2**
15TTF	71.0	**77.0**	74.5	74.9	6.0	75.5
15TUJ	74.2	74.0	**74.5**	73.1	1.4	74.2
15TUH	74.3	76.7	**77.3**	76.8	3.0	77.1
15TUG	75.6	76.1	**77.0**	76.6	1.4	**77.0**
15TUF	72.9	74.7	73.8	**75.1**	2.2	74.6
15TVJ	72.3	75.2	74.6	74.5	3.2	**75.5**
15TVH	76.7	77.5	**78.0**	77.8	1.3	**78.0**
15TVG	68.5	**77.1**	76.9	76.6	8.6	76.6
15TVF	69.8	77.0	**82.1**	77.0	12.3	78.0
15TWJ	61.9	62.0	64.7	**69.3**	7.4	64.8
15TWH	70.3	72.7	71.1	**73.6**	3.3	73.0
15TWG	70.0	**73.4**	71.8	72.9	3.4	72.7
15TWF	79.1	80.3	80.0	80.2	2.0	**81.1**
15TXJ	85.7	85.2	85.4	85.0	0.8	**85.8**
15TXH	74.9	72.3	74.5	77.5	5.8	**78.1**
15TXG	70.9	73.4	**73.6**	72.5	2.7	73.3
15TXF	74.5	75.3	74.8	**75.7**	1.2	75.7
15TXE	77.0	72.5	**77.3**	76.1	5.0	77.5
16TBM	75.8	74.4	74.6	75.3	2.3	**76.7**
Mean	73.6	75.3	75.5	75.8	3.9	**76.2**
Std Dev	4.5	4.1	4.1	**3.2**	2.8	3.8
CV%	6.1	5.6	5.5	**4.3**	71.0	5.2

5.2. Comparison to Cropland Data Layer

For each experiment we reported the overall accuracy, precision (user's accuracy), recall (producer's accuracy), and F1 score for each class in Iowa compared to CDL 2019 (Fig. 7) and Illinois compared to CDL 2020 (Fig. 8). We provided the values used to produce these bar plots and the tile-wise accuracy in Appendix A and Appendix B. The neural network classifiers outperformed the baseline random forest in all metrics and experiments except precision for soybeans using the early-season classifier. Overall, the CNN-LSTM$_\delta$ had better performance for the late-season classification (81% accuracy for Iowa 2019, 85% for Illinois 2020) but was closely followed by the CNN-LSTM (80% Iowa 2019, 85% Illinois 2020) and CNN (80% Iowa 2019, 85% Illinois 2020) models. The CNN had the best performance for the mid-season (80% Iowa 2019, 84% Illinois 2020) and early-season (71% Iowa 2019, 72% Illinois 2020) classifications.

5.3. Comparison to in-season reference dataset

Tables 2–4 give the results for each model in the early-season (greenup timestep only), mid-season (greenup and peak timesteps), and late-season (greenup, peak, and senescence timesteps) classifications. In Table 4, we also reported the performance of the CDL 2020 for the Intelinair reference dataset, although this is not a direct comparison with our results since it is a post-season (not in-season) classification. The relative performance of the compared models based on the Intelinair reference dataset is similar to the CDL test datasets. The CNN-LSTM$_\delta$ had the best overall performance for the late-season classification, closely followed by the CNN and CNN-LSTM, while the CNN had the best overall performance for the mid-season and early-season classifications. The post-season CDL 2020 had comparable performance to our in-season neural network models for the reference dataset.

6. Discussion

6.1. Geographic partitioning of training data

This experiment showed that the universal, cluster, regional, and ensemble models all had higher mean accuracy than the local model. This shows that including training data from other tiles in addition to the test tile improves classification accuracy. This is consistent with the results from Konduri et al. (2020), which showed that optimizing individual models for "ecoregions" determined by clustering had better performance compared to state-level or national-level models (which are comparable to the "regional" and "universal" models in our experiments respectively). While the ensemble model had the highest mean accuracy (as is often the case for ensemble models), the engineering and computational requirements for running the ensemble in an operational setting outweigh the marginal improvement in performance. Of the individual models, the universal model would be the fastest to apply in an operational setting because only one model needs to be trained and the same model is used for inference across all the tiles. In the cluster model, the number of models that need to be trained is equal the number of clusters used (5 in this study), which is much less than the number of tiles. The regional and local models require the same number of models to be trained (one for each tile) and thus are the slowest to apply operationally. Despite greater computational requirements of the regional compared to universal model, this may be a desirable trade-off for some operational use cases. Overall, these results suggest that using a regional approach to selecting model training data will result in models that are more accurate and consistent across tiles compared to the other methods, thus we used this approach for the remaining experiments.

We observed that some tiles had low range of accuracy values across methods while others were quite high, indicating that the choice of model approach may be more important for some regions than others. For example, the difference between the lowest and highest accuracy for 15TXJ is 0.8%, but for 15TVF this difference is 12.3%. These differences in performance across model approaches could be due to local variations in farming practices, weather, distribution of land cover, or other factors.

6.2. Comparison to Cropland Data Layer

In this experiment, we evaluated the proposed methods against the USDA Cropland Data Layer for Iowa in 2019 and Illinois in 2020. The 2019 test set represented an anomalous year in which planting was delayed due to spring flooding while the 2020 test set represented a more standard year. The goal of the proposed phenology normalization is to reduce the impact of non-standard events that occur during the growing season (e.g., late planting) and increase in-season classification errors. Overall, the performance metrics were similar for the two test sets (1–4% difference for best-performing models), but were higher for the Illinois 2020 test set compared to the Iowa 2019 test set. The best-performing method for the early-, mid-, and late-season classifications was also the same for both datasets: CNN-LSTM$_\delta$ for late-season and CNN for mid- and early-season classifications. This suggests that the δ feature is more useful for discriminating between corn and soybean samples when it is a measure of the number of days between the greenup and senescence stages than between the peak and senescence stages. In the early-season classification, the higher performance of the CNN compared to the random forest suggests that the model may be learning spatial or textural features that are useful for classification in the early season.

6.3. Comparison to in-season reference dataset

In this experiment, we evaluated the proposed methods against a test set of 3973 crop type reference labels collected during the growing season by Intelinair. For all models, the performance for the "other"

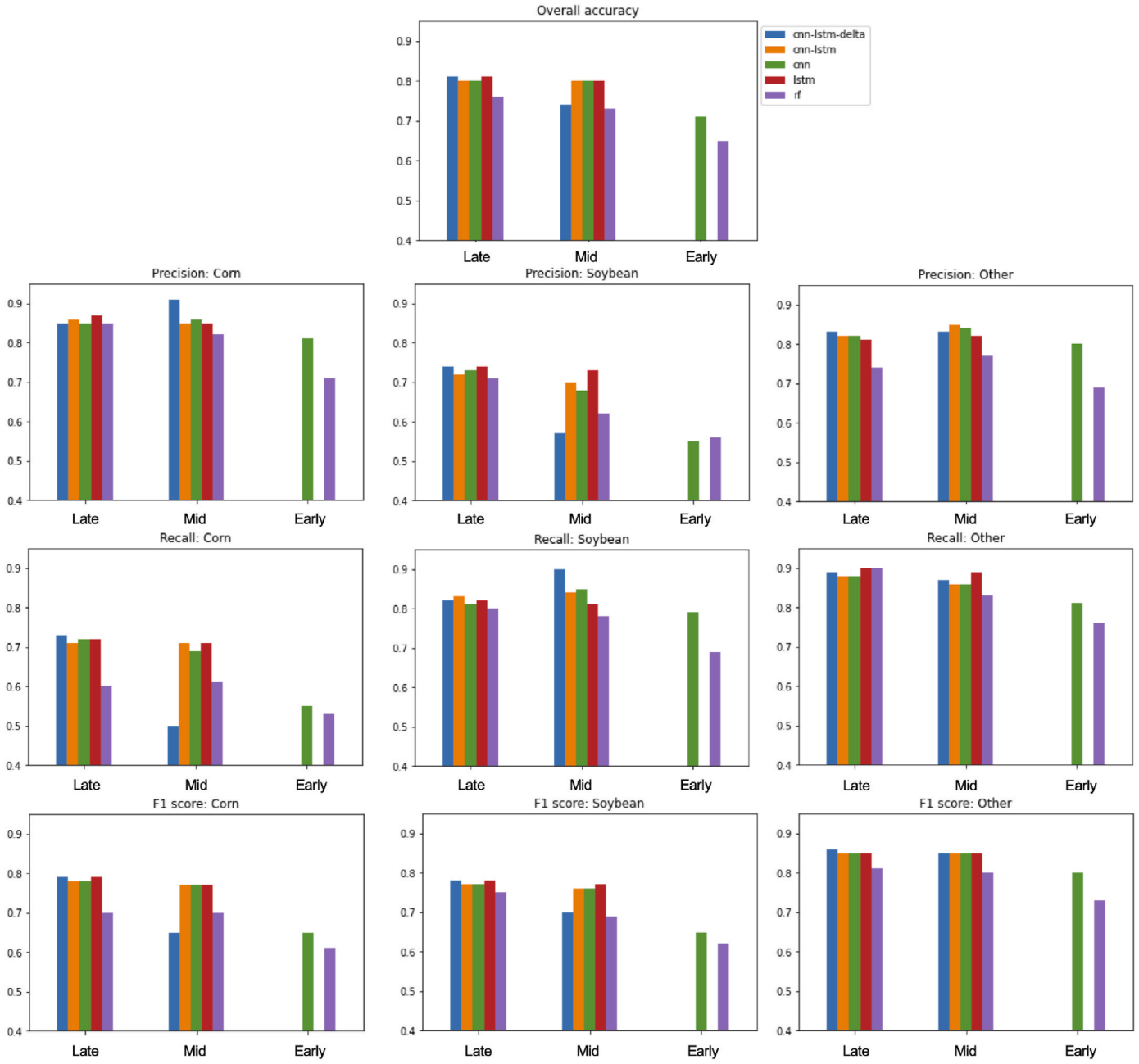


Fig. 7. Overall accuracy and per-class precision, recall, and F1 score for CDL 2019 test set (Iowa).

class was substantially lower than in the CDL experiments, likely because the "other" class from CDL includes non-crop land cover classes (e.g., water or forest) in addition to other crop types while the Intelinair dataset only includes other crop types. It is interesting that, even though our models were trained using labels from CDL 2017–2019, our models in some cases have higher metrics than the CDL 2020 (Table 4). In addition, comparing the CDL column from Table 4 to Table 3, we observe that our CNN model achieves comparable performance in mid-season to the CDL 2020, which is a post-season classification, for this dataset.

Since crop type classification models can be trained using labels from prior seasons, in-season labels are not required for predicting in-season crop type maps. However, some in-season labels are still needed for validating the predicted in-season crop type maps, especially if the maps will be used for informing policy or decision-making. In future work, we will continue to explore strategies to obtaining larger ground-truth or reference datasets, including collaboration with private companies like Intelinair and crowdsourcing approaches.

6.4. *Potential sources of error*

These results could be improved in future work by improving the consistency and accuracy of the phenology detection method. Noise in the time series (e.g., due to clouds not detected by the cloud mask) could result in spurious detections. This could be addressed by using smoothing methods instead of/in addition to linear interpolation or improved cloud masking/outlier removal. However, smoothing can add considerable computational time since they must be applied for every pixel, thus future work could use larger units of classification (e.g., objects or parcels) to speed up computation. Another source of error in the phenology detection could be due to differences in planting timelines across the study regions and US Midwest more broadly. Currently, our

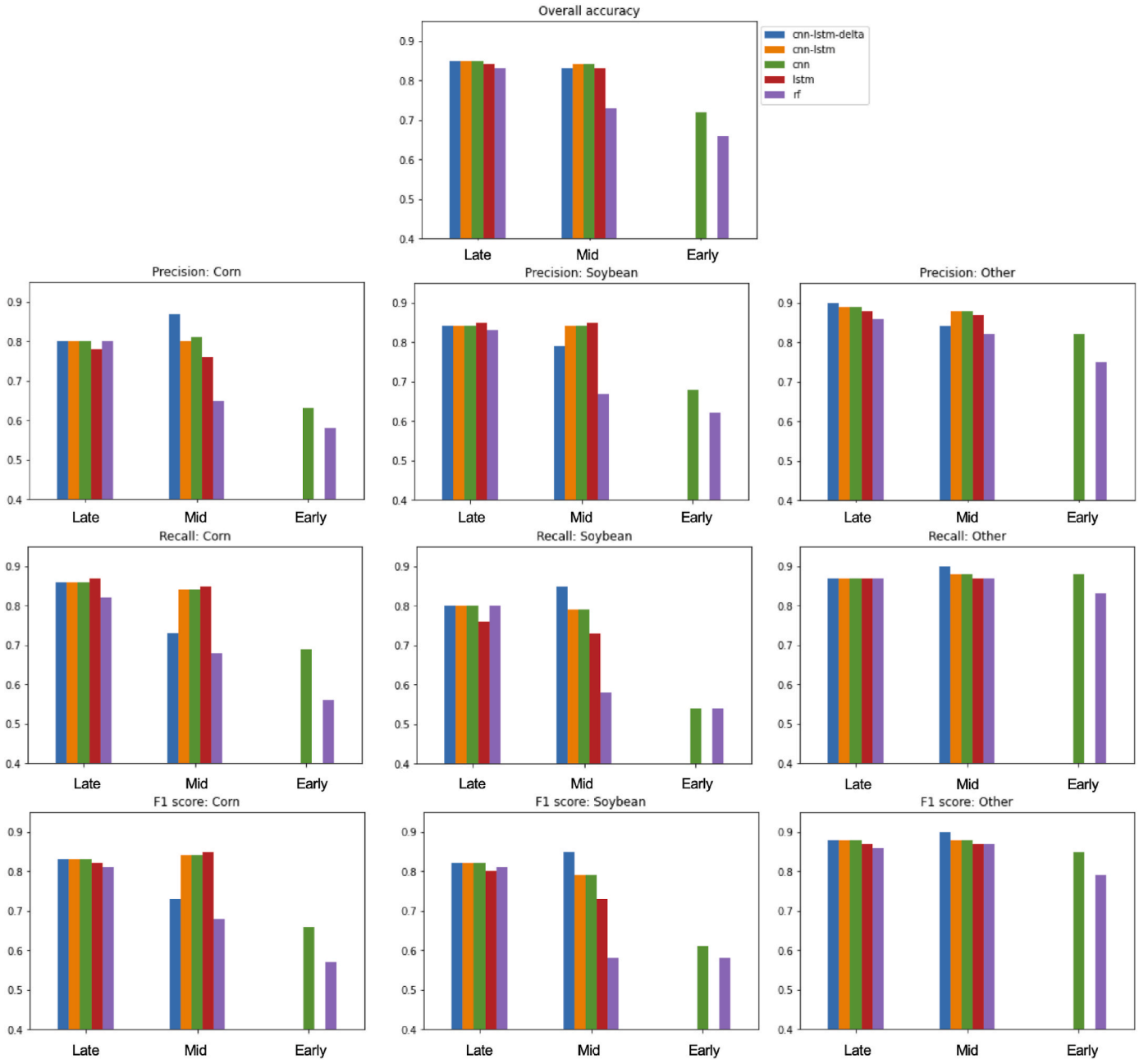


Fig. 8. Overall accuracy and per-class precision, recall, and F1 score for CDL 2020 test set (Illinois).

Table 2
Performance metrics for early-season (1-timestep) classification experiments, using Intelinair 2020 reference labels. **Bold** indicates best performance.

Metric	CNN	Random Forest
Overall accuracy	**0.71**	0.62
Precision (UA)		
Corn	**0.81**	0.80
Soybean	**0.70**	0.57
Other	**0.23**	0.18
Recall (PA)		
Corn	**0.76**	0.62
Soybean	**0.62**	0.61
Other	0.60	**0.61**
F1 score		
Corn	**0.79**	0.70
Soybean	**0.66**	0.59
Other	**0.34**	0.28

method only considers the time period between April 15 through July 1 for greenup detection, but soybean may be planted in late June in some places in the Corn Belt which would make the greatest increase in greenness much later than July 1. Future work could consider defining an adaptive window for detection based on local crop calendars.

Inaccuracies in the USDA Cropland Data Layer products used for training and evaluating models is another potential source of error. We tried to remove samples that could have errors from the training set by aggregating crop type classes and only sampling pixels from homogeneous regions of the map, but this is not robust to all errors and could have the negative effect of removing some diversity from the training set. Other studies have employed these same approaches to reduce CDL errors Wang et al. (2020); Gray et al. (2021), as well as others such as selecting only those samples that are consistent with well-known crop rotations Zhang et al. (2021). CDL errors could also affect the interpretation of the results for the experiments in which the CDL was used as

Table 3
Performance metrics for mid-season (2-timestep) classification experiments, using Intelinair 2020 reference labels. **Bold** indicates best performance.

Metric	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Overall accuracy	0.86	0.88	**0.89**	0.86	0.69
Precision (UA)					
Corn	**0.96**	0.93	0.95	0.90	0.82
Soybean	0.85	**0.90**	0.89	**0.90**	0.65
Other	0.36	0.40	**0.42**	0.38	0.24
Recall (PA)					
Corn	0.86	**0.91**	0.90	0.90	0.73
Soybean	**0.89**	0.85	0.88	0.79	0.62
Other	**0.76**	0.72	0.72	0.71	0.66
F1 score					
Corn	0.91	**0.92**	**0.92**	0.90	0.77
Soybean	0.87	0.88	**0.89**	0.84	0.64
Other	0.49	0.51	**0.53**	0.50	0.35

Table 4
Performance metrics for late-season (3-timestep) classification experiments, using Intelinair 2020 reference labels. We also report the performance for the CDL 2020 compared to the reference test set, even though it is not an in-season classification and is not directly comparable to our methods. **Bold** indicates best performance (excluding CDL).

Metric	CDL	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Overall accuracy	0.89	**0.89**	**0.89**	**0.89**	0.88	0.87
Precision (UA)						
Corn	0.96	0.93	0.93	**0.94**	0.93	**0.94**
Soybean	0.89	**0.90**	**0.90**	0.88	**0.90**	0.88
Other	0.34	**0.50**	0.45	**0.50**	0.40	0.35
Recall (PA)						
Corn	0.91	**0.93**	0.92	0.92	0.91	0.89
Soybean	0.91	0.86	0.86	**0.87**	0.84	0.86
Other	0.54	0.68	0.68	**0.70**	0.66	0.62
F1 score						
Corn	0.93	**0.93**	**0.93**	**0.93**	0.92	0.91
Soybean	0.90	**0.88**	**0.88**	**0.88**	0.87	0.87
Other	0.41	**0.58**	0.55	**0.58**	0.50	0.45

the test set, which did not have any filters applied.

Land cover or crop type classification models generally have higher performance for discriminating between classes that exhibit very different land surface phenology (e.g., corn versus wheat). While phenology normalization improves the generalization of classifiers to seasons that exhibit patterns unseen in training data collected in prior years, one potential limitation is that normalization could remove features related to differences in land surface phenology observed for different crops that would otherwise be useful for the classifier to distinguish between crop types.

6.5. Summary of findings in response to research questions

In this study, we presented an approach for predicting crop type maps for operational applications using moderate resolution (30 m) satellite observations, including Harmonized Landsat and Sentinel-2 (HLS) optical and Sentinel-1 SAR data. Here we summarize our findings for the research questions outlined in Section 1:

- What is the best approach for selecting training data in terms of geographic diversity for operational crop mapping using moderate (30 m) satellite imagery?

We evaluated four approaches to selecting training data (Fig. 6) in which training samples are drawn from the same tile as the test tile (local model), from the same tile and tiles adjacent to the test tile (regional model), from pixels belonging to the same cluster as the test pixels (cluster model), and from all tiles in the study area (universal model). We found that the regional model had the best performance in terms of accuracy and variability across test tiles of the four approaches (Table 1). We found that slightly higher accuracy could be achieved using an ensemble of all four approaches, but the marginal improvement (0.5%) is likely not worth the added computational complexity for most operational use cases.

- How can we prevent reduced crop type map accuracy for unseen years where there is a domain shift with respect to prior years (e.g., due to late planting)?

We proposed an approach that we refer to as *phenology normalization* to construct a series of inputs based on land surface phenology stage in each pixel rather than absolute time. In each pixel, we find the approximate date at which the greenup, peak, and senescence land surface phenology stages are reached, then select the HLS and Sentinel-1 values at those dates. Thus each image in the series is a composite in which all pixels are at approximately the same phenology stage. This method can be used for in-season classification at 3 stages: early-season (only greenup input used), mid-season (greenup and peak inputs), and late-season (greenup, peak, and senescence inputs). Using test data from 2019 (an anomalous year due to widespread late planting in the mid-western US) and 2020 (a standard planting year), we showed that this method enables comparable performance for both years (81% overall accuracy for Iowa 2019, 85% Illinois 2020 compared to CDL).

- Which crop type classification methods have the best performance throughout the growing season (e.g., early, mid, and late season)?

Our proposed classification architecture extracts temporal features (using an LSTM) from the pixel-level time series and spatial features (using a CNN) from a small patch around the pixel, which are combined to make a pixel-level prediction of three output classes (corn, soybean, or other). When 2 or more timesteps are used, we include an additional feature that is the difference (in days) between the first and last timesteps. We also evaluated performance when using only the CNN or LSTM component as well as a random forest baseline. We found that for the late-season classification, the CNN-LSTM$_\delta$ had the best overall performance (81% for Iowa 2019, 85% for Illinois 2020), while the CNN had better performance for the early- (71% for Iowa 2019, 72% for Illinois 2020) and mid-season (80% for Iowa 2019, 84% for Illinois 2020) classifications. We found that while the random forest had similar performance for the late-season classification to the neural network methods, it had much lower performance for the early- and mid-season classifications. We used CDL as a surrogate for ground-truth reference labels in both training and evaluation, thus the training and test labels include some mislabeled points, which affect the assessment of model performance. We also evaluated the models against a reference test set provided by Intelinair including 3973 fields in two states in the US Midwest. All models had higher accuracy for this dataset (e.g., 89% for late-season using CNN-LSTM$_\delta$) compared to the CDL test sets, though we note that these samples are sparse compared to the CDL and do not include non-crop classes in the "other" class as in the CDL.

7. Conclusions

Machine learning and remote sensing data can be used to create crop type maps that provide classifications of crop-specific land cover to enable timely analysis at field scales throughout the growing season to supplement traditional data collection and reporting efforts. In this study, we presented an approach for predicting crop type maps for operational applications using moderate resolution (30 m) satellite observations using HLS optical and Sentinel-1 SAR data. A key component of this approach is the application of *phenology normalization* prior to

classification, which tries to improve model generalization for future unseen years that may exhibit inter-annual domain shifts due to, e.g., late planting. The phenology-normalized observations are then used as input to a neural network that uses recurrent and convolutional layers to extract temporal and spatial features, respectively. This approach can be used for in-season classification by including only the inputs for the phenology stages that have been observed at a given point in the growing season (e.g., only the greenup input is used in early-season). We evaluated our approach using the USDA Cropland Data Layer in Iowa in 2019 (an anomalous year due to late planting) and Illinois in 2020 (representing a typical year). We found that the CNN-LSTM$_\delta$ had the best overall performance for late-season classification (81% for Iowa 2019, 85% for Illinois 2020), while the CNN had better performance for early-season (71% for Iowa 2019, 72% for Illinois 2020) and mid-season (80% for Iowa 2019, 84% for Illinois 2020) classifications. We also evaluated model performance using a reference dataset of crop type in 3973 fields. The highest overall accuracy for this test set was 89% for late-season classification (CNN-LSTM$_\delta$, CNN-LSTM, and CNN), 89% for mid-season (CNN), and 71% for early-season (CNN). We performed an experiment to evaluate different model training approaches based on geographic partitioning of the training data. We found that the regional model—in which one model is trained for each tile using training samples from that tile and adjacent tiles—had the best overall performance, though with a tradeoff for higher computational costs since it requires training a separate classifier for every tile.

In future work, we will apply this approach to generate publicly-available in-season crop type maps for future growing seasons operationally. We will continue to investigate methods for improving performance of the land surface phenology normalization and crop type classification components of the approach.

Declaration of interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hannah Kerner reports financial support was provided by NASA. Inbal Becker-Reshef reports financial support was provided by NASA.

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Appendix A. Detailed results for CDL 2019 (Iowa) test set

Table A.5
Performance metrics for late-season (3-timestep) classification experiments, using CDL 2019 for reference labels. Mean reported across all HLS tiles overlapping with Iowa (see A.8 for per-tile accuracy). **Bold** indicates best performance.

Metric	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Overall accuracy	**0.81**	0.80	0.80	**0.81**	0.76
Precision (UA)					
Corn	0.85	0.86	0.85	**0.87**	0.85
Soybean	**0.74**	0.72	0.73	**0.74**	0.71
Other	**0.83**	0.82	0.82	0.81	0.74
Recall (PA)					
Corn	**0.73**	0.71	0.72	0.72	0.60
Soybean	0.82	**0.83**	0.81	0.82	0.80
Other	0.89	0.88	0.88	**0.90**	**0.90**
F1 score					
Corn	**0.79**	0.78	0.78	**0.79**	0.70
Soybean	**0.78**	0.77	0.77	**0.78**	0.75
Other	**0.86**	0.85	0.85	0.85	0.81

Table A.6
Performance metrics for mid-season (2-timestep) classification experiments, using CDL 2019 for reference labels. Mean reported across all HLS tiles overlapping with Iowa (see A.9 for per-tile accuracy). **Bold** indicates best performance.

Metric	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Overall accuracy	0.74	**0.80**	**0.80**	**0.80**	0.73
Precision (UA)					
Corn	**0.91**	0.85	0.86	0.85	0.82
Soybean	0.57	0.70	0.68	**0.73**	0.62
Other	0.83	**0.85**	0.84	0.82	0.77
Recall (PA)					
Corn	0.50	**0.71**	0.69	**0.71**	0.61
Soybean	**0.90**	0.84	0.85	0.81	0.78
Other	0.87	0.86	0.86	**0.89**	0.83
F1 score					
Corn	0.65	**0.77**	**0.77**	**0.77**	0.70
Soybean	0.70	0.76	0.76	**0.77**	0.69
Other	**0.85**	**0.85**	**0.85**	**0.85**	0.80

Table A.7
Performance metrics (overall accuracyand per-class precision/user's accuracy (UA), recall/producer's accuracy (PA), and F1 score) for early-season (1-timestep) classification experiments, using CDL 2019 for reference labels. Mean reported across all HLS tiles overlapping with Iowa (see A.10 for per-tile accuracy). **Bold** indicates best performance.

Metric	CNN	Random Forest
Overall accuracy	**0.71**	0.65
Precision (UA)		
Corn	**0.81**	0.71
Soybean	0.55	**0.56**
Other	**0.80**	0.69
Recall (PA)		
Corn	**0.55**	0.53
Soybean	**0.79**	0.69
Other	**0.81**	0.76
F1 score		
Corn	**0.65**	0.61
Soybean	**0.65**	0.62
Other	**0.80**	0.73

Table A.8
Overall accuracy for late-season (3-timestep) classification experiments, using CDL 2019 for reference labels (Iowa). **Bold** indicates best performance. Range column indicates difference between maximum and minimum score across the five methods for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile ID	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest	Range
14TPP	80.1	79.0	73.2	80.7	**82.3**	9.1
14TPN	78.9	79.2	**80.0**	79.7	79.2	1.1
14TQN	79.6	79.8	80.4	80.6	**81.5**	1.9
14TQP	64.0	65.9	**70.7**	68.2	65.3	6.7
15TTG	**79.8**	79.7	78.1	77.3	78.1	2.5
15TTF	77.4	77.2	80.2	77.3	**78.8**	3.0
15TUJ	**82.0**	80.2	76.8	79.9	72.4	9.6
15TUH	**84.2**	82.9	81.4	81.2	76.8	7.4
15TUG	**83.3**	82.4	82.0	81.0	78.4	4.9
15TUF	**80.7**	79.6	81.2	79.3	77.8	3.4
15TVJ	**79.8**	76.7	78.9	76.2	73.1	6.7
15TVH	83.4	**84.1**	83.3	81.0	78.4	5.7
15TVG	**83.6**	82.6	78.6	81.5	78.7	5.0
15TVF	**87.1**	86.0	85.6	81.7	82.1	5.4
15TWJ	**77.2**	76.5	75.9	71.7	69.7	7.5
15TWH	78.1	**78.2**	77.8	76.2	70.3	7.9
15TWG	78.8	78.6	**79.3**	78.7	70.9	8.4
15TWF	**82.1**	80.9	81.6	80.9	79.6	2.5
15TXJ	**89.8**	89.2	88.9	87.1	87.4	2.7
15TXH	**84.3**	82.5	81.9	80.2	76.2	8.1
15TXG	**76.8**	74.3	77.6	75.8	68.1	9.5
15TXF	**77.7**	75.3	74.5	74.0	68.0	9.7
15TXE	81.8	82.3	**82.4**	79.2	75.9	6.5
16TBM	71.1	69.8	72.2	**74.5**	64.3	10.2
Mean	**80.1**	79.3	79.3	78.5	75.5	6.1
Std Dev	5.0	4.8	4.1	**3.8**	5.7	2.7
CV%	6.3	6.0	5.1	**4.7**	7.1	45.2

Table A.9
Overall accuracy for mid-season (2-timestep) classification experiments, using CDL 2019 for reference labels (Iowa). **Bold** indicates best performance. Range column indicates difference between maximum and minimum score across the five methods for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile ID	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest	Range
14TPP	73.1	78.1	77.3	**80.3**	64.2	16.1
14TPN	73.9	81.1	**82.0**	81.2	70.0	12.0
14TQN	69.1	78.0	77.5	**80.8**	74.7	11.7
14TQP	62.1	70.9	**73.5**	70.0	70.2	11.4
15TTG	69.1	**78.9**	77.0	77.8	70.7	9.8
15TTF	68.1	**76.7**	76.3	76.2	73.2	8.6
15TUJ	67.0	**76.5**	75.5	77.4	72.0	10.4
15TUH	72.5	81.6	**81.8**	79.2	76.6	9.3

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Table A.9 (*continued*)

Tile ID	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest	Range
15TUG	76.0	**83.0**	82.3	81.3	76.4	7.0
15TUF	73.0	79.2	**80.2**	79.4	72.9	7.3
15TVJ	76.8	80.0	**80.1**	74.2	70.7	9.4
15TVH	72.1	**82.5**	81.1	79.2	77.0	10.4
15TVG	67.6	79.8	78.5	**78.7**	74.9	12.2
15TVF	82.0	**83.2**	82.0	77.7	71.5	11.7
15TWJ	72.0	**74.2**	72.5	71.4	63.1	11.1
15TWH	72.8	75.5	**77.5**	73.5	66.1	11.4
15TWG	70.8	77.4	77.4	**77.5**	69.5	8.0
15TWF	74.3	**81.1**	80.9	80.0	75.6	6.8
15TXJ	**87.4**	86.9	86.9	85.6	75.8	11.6
15TXH	81.2	**83.2**	82.3	80.4	72.7	10.5
15TXG	72.6	**75.9**	75.3	73.9	67.2	8.7
15TXF	71.8	74.5	**75.1**	74.2	69.6	5.5
15TXE	78.0	79.4	**80.3**	79.9	68.0	12.3
16TBM	68.6	71.6	**72.3**	68.7	65.7	6.6
Mean	73.0	**78.7**	78.6	77.4	71.2	10.0
Std Dev	5.3	3.8	**3.5**	3.9	3.9	2.3
CV%	7.3	5.2	**4.8**	5.4	5.4	23.5

Table A.10
Overall accuracy for early-season (1-timestep) classification experiments, using CDL 2019 for reference labels (Iowa). We did not evaluate the models involving the LSTM since the input contains a single timestep. **Bold** indicates best performance. Range column indicates difference between the two scores for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile ID	CNN	Random Forest	Range
14TPP	**61.2**	54.0	7.2
14TPN	**61.8**	61.1	0.7
14TQN	**62.6**	60.7	1.9
14TQP	**71.6**	62.0	9.6
15TTG	**65.6**	59.2	6.4
15TTF	**67.8**	67.3	0.5
15TUJ	**66.0**	62.8	3.2
15TUH	**74.0**	69.2	4.8
15TUG	**76.1**	68.0	8.1
15TUF	**73.0**	68.1	4.9
15TVJ	**67.4**	61.4	6.0
15TVH	**76.5**	68.7	7.8
15TVG	**69.8**	66.4	3.4
15TVF	**71.3**	64.2	7.1
15TWJ	**65.2**	51.2	14.0
15TWH	**68.8**	60.5	8.3
15TWG	**70.3**	61.8	8.5
15TWF	**74.7**	71.5	3.2
15TXJ	**84.1**	65.8	18.3
15TXH	**74.0**	60.4	13.6
15TXG	**65.5**	61.9	3.6
15TXF	62.3	**63.8**	1.5
15TXE	**68.9**	61.4	7.5
16TBM	54.7	**57.2**	2.5
Mean	**68.9**	62.9	6.4
Std Dev	6.1	**4.7**	4.3
CV%	8.9	**6.8**	67.8

Appendix B. Detailed results for CDL 2020 (Illinois) test set

Table B.11
Performance metrics for late-season (3-timestep) classification experiments, using CDL 2020 for reference labels. Mean reported across all HLS tiles overlapping with Illinois (see B.14 for per-tile accuracy). **Bold** indicates best performance.

Metric	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Overall accuracy	**0.85**	**0.85**	**0.85**	0.84	0.83
Precision (UA)					
Corn	**0.80**	**0.80**	**0.80**	0.78	**0.80**
Soybean	0.84	0.84	0.84	**0.85**	0.83
Other	**0.90**	0.89	0.89	0.88	0.86

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Table B.11 (*continued*)

Metric	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Recall (PA)					
Corn	0.86	0.86	0.86	**0.87**	0.82
Soybean	**0.80**	**0.80**	**0.80**	0.76	**0.80**
Other	**0.87**	**0.87**	**0.87**	**0.87**	**0.87**
F1 score					
Corn	**0.83**	**0.83**	**0.83**	0.82	0.81
Soybean	**0.82**	**0.82**	**0.82**	0.80	0.81
Other	**0.88**	**0.88**	**0.88**	0.87	0.86

Table B.12
Performance metrics for mid-season (2-timestep) classification experiments, using CDL 2020 for reference labels. Mean reported across all HLS tiles overlapping with Illinois (see B.15 for per-tile accuracy). **Bold** indicates best performance.

Metric	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest
Overall accuracy	0.83	**0.84**	**0.84**	0.83	0.73
Precision (UA)					
Corn	**0.87**	0.80	0.81	0.76	0.65
Soybean	0.79	0.84	0.84	**0.85**	0.67
Other	0.84	**0.88**	**0.88**	0.87	0.82
Recall (PA)					
Corn	0.73	0.84	0.84	**0.85**	0.68
Soybean	**0.85**	0.79	0.79	0.73	0.58
Other	**0.90**	0.88	0.88	0.87	0.87
F1 score					
Corn	0.79	**0.82**	**0.82**	0.80	0.67
Soybean	**0.82**	0.81	**0.82**	0.79	0.62
Other	0.87	**0.88**	**0.88**	0.87	0.84

Table B.13
Performance metrics (overall accuracy, per-class precision/user's accuracy (UA) and recall/producer's accuracy (PA)) for each method for early-season (1-timestep) classification experiments, using CDL 2020 for reference labels. Mean reported across all HLS tiles overlapping with Illinois (see B.16 for per-tile accuracy). **Bold** indicates best performance for each experiment.

Metric	CNN	Random Forest
Overall accuracy	**0.72**	0.66
Precision (UA)		
Corn	**0.63**	0.58
Soybean	**0.68**	0.62
Other	**0.82**	0.75
Recall (PA)		
Corn	**0.69**	0.56
Soybean	**0.54**	**0.54**
Other	**0.88**	0.83
F1 score		
Corn	**0.66**	0.57
Soybean	**0.61**	0.58
Other	**0.85**	0.79

Table B.14
Overall accuracy for each method for late-season (3-timestep) classification experiments, using CDL 2020 for reference labels (Illinois). **Bold** indicates best performance. Range column indicates difference between maximum and minimum score across the five methods for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile ID	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest	Range
16TBK	**87.3**	87.2	86.8	86.0	86.5	1.3
16SBJ	**87.5**	87.3	87.3	85.8	86.5	1.7
16TCK	84.1	**85.7**	84.6	84.2	81.0	4.7
16SCJ	82.6	82.6	**83.7**	80.7	82.7	3.0
16SCH	82.4	82.1	**82.8**	79.1	80.5	3.7
16TBL	**88.4**	87.8	88.1	86.1	86.8	2.3
16TDK	84.4	84.0	**85.3**	83.1	83.4	2.2
16TDL	**79.0**	77.6	77.3	74.8	76.9	4.2

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Table B.14 (*continued*)

Tile ID	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest	Range
16SDJ	**83.0**	82.0	82.2	82.0	**83.0**	1.0
15TYE	87.1	**87.5**	86.8	86.5	86.7	1.0
15TXF	**86.8**	86.0	85.8	85.5	84.5	2.3
15TXE	**88.0**	87.9	**88.0**	86.9	87.6	1.1
15TXH	82.2	82.8	77.1	82.6	**82.9**	5.8
16TBM	**86.8**	86.4	86.3	83.7	84.8	3.1
15TYH	**85.9**	84.7	85.0	85.4	85.3	1.2
15SYD	**88.0**	87.7	87.8	84.1	87.1	3.9
16TCN	**83.1**	82.3	82.7	79.5	82.1	3.6
16TDN	96.8	**96.9**	96.8	96.4	96.4	0.5
16TCM	83.5	82.6	82.9	82.4	**83.9**	1.5
16TDM	97.9	**98.0**	97.7	96.2	97.7	1.8
16TCL	80.8	80.3	**81.1**	78.3	76.6	4.5
16SBH	**87.6**	87.3	87.0	85.7	85.5	2.1
16SDG	88.9	88.7	**89.0**	86.4	87.5	2.6
16SCG	88.9	88.9	**89.5**	86.3	87.0	3.2
16SBG	**90.0**	89.7	**90.0**	86.8	88.7	3.2
Mean	**86.3**	86.1	86.0	84.5	85.1	2.6
Std Dev	**4.2**	4.4	4.6	4.5	4.6	1.3
CV%	**4.9**	5.1	5.3	5.3	5.3	50.5

Table B.15
Overall accuracy for each method for mid-season (2-timestep) classification experiments, using CDL 2020 for reference labels (Illinois). **Bold** indicates best performance. Range column indicates difference between maximum and minimum score across the five methods for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile ID	CNN-LSTM$_\delta$	CNN-LSTM	CNN	LSTM	Random Forest	Range
16TBK	**86.8**	86.4	86.7	84.8	74.4	12.4
16SBJ	85.7	86.2	**86.3**	85.0	73.8	12.5
16TCK	85.8	**86.3**	84.9	81.7	65.5	20.8
16SCJ	83.0	**83.6**	83.5	80.4	67.7	15.9
16SCH	82.6	81.5	**82.7**	78.7	67.4	15.3
16TBL	87.0	86.5	**87.3**	85.3	77.5	9.8
16TDK	82.5	**84.5**	83.8	81.2	66.2	18.3
16TDL	73.6	**79.2**	78.0	72.6	66.5	12.7
16SDJ	81.9	82.0	**82.7**	81.3	69.2	13.5
16SDH	**82.9**	82.5	81.3	80.1	67.4	15.5
15TYE	86.3	86.3	**86.5**	85.2	76.2	10.3
15TXF	83.7	**84.1**	83.9	82.2	77.2	6.9
15TXE	**86.8**	86.4	86.3	85.3	77.9	8.9
15TXH	82.9	82.9	**83.0**	81.0	75.5	7.5
16TBM	85.2	85.3	**85.4**	82.8	71.8	13.6
15TYH	**85.5**	85.0	84.5	82.5	76.9	8.6
15SYD	86.2	**86.6**	86.2	81.9	76.2	10.4
16TCN	81.4	**82.0**	81.3	79.9	71.9	10.1
16TDN	**96.8**	96.7	**96.8**	95.8	93.3	3.5
16TCM	78.2	82.2	**84.0**	82.2	71.9	12.1
16TDM	**98.0**	97.8	97.8	96.4	91.6	6.4
16TCL	73.8	78.3	**79.0**	77.8	70.2	8.8
16SBH	86.6	**87.2**	87.0	84.8	77.5	9.7
16SDG	**88.7**	87.8	88.1	85.8	75.9	12.8
16SCG	**88.7**	87.8	88.1	85.3	77.9	10.8
16SBG	**90.1**	90.0	90.0	86.3	81.7	8.4
Mean	85.0	**85.6**	**85.6**	83.3	74.6	11.4
Std Dev	5.3	**4.3**	**4.3**	4.7	6.8	3.8
CV%	6.2	**5.1**	**5.1**	5.6	8.0	33.0

Table B.16
Overall accuracy for each method for early-season (1-timestep) classification experiments, using CDL 2020 for reference labels (Illinois). **Bold** indicates best performance. Range column indicates difference between the two scores for each row. Mean, standard deviation, and coefficient of variation (CV%) reported for each column.

Tile ID	CNN	Random Forest	Range
16TBK	**72.8**	66.4	6.4
16SBJ	**74.4**	67.9	6.5
16TCK	**70.3**	63.0	7.3

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Table B.16 (*continued*)

Tile ID	CNN	Random Forest	Range
16SCJ	**71.0**	63.6	7.4
16SCH	**67.7**	61.4	6.3
16TBL	**76.4**	64.0	12.4
16TDK	**69.6**	61.4	8.2
16TDL	**62.0**	61.3	0.7
16SDJ	**71.0**	65.6	5.4
16SDH	**69.1**	59.4	9.7
15TYE	**74.0**	68.2	5.8
15TXF	**74.7**	63.4	11.3
15TXE	**76.5**	70.4	6.1
15TXH	**70.0**	67.5	2.5
16TBM	**70.6**	62.6	8.0
15TYH	**76.0**	72.5	3.5
15SYD	**77.2**	69.2	8.0
16TCN	**69.5**	66.8	2.7
16TDN	**94.3**	93.1	1.2
16TCM	**67.7**	64.2	3.5
16TDM	**95.2**	92.1	3.1
16TCL	**64.6**	63.5	1.1
16SBH	**79.1**	70.6	8.5
16SDG	**79.1**	69.2	9.9
16SCG	**80.1**	70.8	9.3
16SBG	**85.3**	76.7	8.6
Mean	**74.5**	68.3	6.3
Std Dev	**7.7**	8.1	3.1
CV%	**10.3**	10.8	50.0

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